

Hundred days of Coding in C Programming

Day-19

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Batch - 15

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Question

The algorithm for calculating the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) for 'm' semesters is given below. Each semester has 'n' courses, where 'n' may vary between semesters. Assume that the student has passed all courses.

1. Prepare a test-case table covering valid, invalid, and boundary inputs.
2. Write the C program based on the given algorithm.
3. Compile and execute it.
4. Test it against the prepared test cases.

Given Algorithm

Start.

Define the variables:

- `global_id`, `m`, `n`, `semester`, `course`, `code`, `credit`, `GP`
- `semester_TC`, `semester_TGC`, `cumulative_TC`, `cumulative_TGC` as integers.
- `grade` as a character.
- `SGPA` and `CGPA` as floating-point variables.

Get `global_id`.

Validate `global_id`. If it is not positive, display an error message and request it again until a positive value is entered.

Get `m`.

Validate `m`. If it is not positive, display an error message and request it again until a positive value is entered.

Set:

- `cumulative_TC = 0`
- `cumulative_TGC = 0`

Repeat the following steps for **m semesters**:

- Set `semester_TC = 0` and `semester_TGC = 0`.
- Get `n`.
- Validate `n`. If it is not positive, request it again.
- Repeat the following steps for **n courses**:
 - Get `code`.
 - Validate `code`. It must be positive.

- Get `credit`.
- Validate `credit`. It must be positive.
- Get `grade`.
- Validate grade. Accept only **O, A, B, C, D, E, or P**.
- Find GP according to the conversion scale:
 - O = 10
 - A = 9
 - B = 8
 - C = 7
 - D = 6
 - E = 5
 - P = 4
- Calculate:

$$\text{semester_TGC} = \text{semester_TGC} + (\text{GP} \times \text{credit})$$
- Calculate:

$$\text{semester_TC} = \text{semester_TC} + \text{credit}$$
- Calculate:

$$\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$$
- Display semester and SGPA.
- Update:

$$\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$$

$$\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$$

Calculate:

$$\text{CGPA} = \text{cumulative_TGC} / \text{cumulative_TC}$$

Display `global_id`, `m`, `cumulative_TC`, and CGPA.

Stop.

Step - 1 Algorithm

1. Start.
2. Enter and validate Global ID.
3. Enter and validate number of semesters (m).
4. Set cumulative credits and cumulative grade credits to 0.
5. For each semester, enter and validate the number of courses (n).
6. For each course, enter course code, credit and grade and validate them.
7. Convert grade into grade point: O=10, A=9, B=8, C=7, D=6, E=5, P=4.
8. Calculate semester total credits and total grade credits.
9. Calculate SGPA = semester grade credits / semester credits.
10. Add semester values to cumulative values.
11. Calculate CGPA = cumulative grade credits / cumulative credits.
12. Display the result.
13. Stop.

Step - 2 Grade Point Table

Grade	GP
O	10
A	9
B	8
C	7
D	6
E	5
P	4

Step - 3 Test Case Table

Test Case	Input	Type	Expected Result
1	ID=101, m=1, n=2, credits=3,3,	Valid	SGPA = 8.50, CGPA = 8.50
2	Global ID = -1	Invalid	Error message; enter ID again
3	Number of semesters = 0	Invalid	Error message; enter value again
4	Credit = -2	Invalid	Error message; enter credit again
5	Grade = X	Invalid	Error message; enter valid grade
6	ID=1, m=1, n=1, credit=1,	Boundary	SGPA = 10.00, CGPA = 10.00

Step - 4 C Program

```
#include <stdio.h>
```

```
int main()
{
    int global_id, m, n;
    int semester, course, code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    do {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0)
            printf("Invalid Global ID!\n");
    } while (global_id <= 0);

    do {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0)
            printf("Invalid number of semesters!\n");
    } while (m <= 0);

    for (semester = 1; semester <= m; semester++)
    {
        semester_TC = 0;
        semester_TGC = 0;

        do {
            printf("\nEnter number of courses for Semester %d: ", semester);
            scanf("%d", &n);
            if (n <= 0)
                printf("Invalid number of courses!\n");
        } while (n <= 0);

        for (course = 1; course <= n; course++)
        {
            do {
                printf("Enter course code: ");
                scanf("%d", &code);
                if (code <= 0)
                    printf("Invalid course code!\n");
            } while (code <= 0);

            do {
                printf("Enter credit: ");
                scanf("%d", &credit);
                if (credit <= 0)
                    printf("Invalid credit!\n");
            } while (credit <= 0);

            do {
                printf("Enter grade (O/A/B/C/D/E/P): ");
                scanf(" %c", &grade);

                if (grade != 'O' && grade != 'A' && grade != 'B' &&
                    grade != 'C' && grade != 'D' && grade != 'E' &&
```

```

        grade != 'P')
        printf("Invalid grade!\n");

    } while (grade != 'O' && grade != 'A' && grade != 'B' &&
            grade != 'C' && grade != 'D' && grade != 'E' &&
            grade != 'P');

    if (grade == 'O')
        GP = 10;
    else if (grade == 'A')
        GP = 9;
    else if (grade == 'B')
        GP = 8;
    else if (grade == 'C')
        GP = 7;
    else if (grade == 'D')
        GP = 6;
    else if (grade == 'E')
        GP = 5;
    else
        GP = 4;

    semester_TGC += GP * credit;
    semester_TC += credit;
}

SGPA = (float)semester_TGC / semester_TC;
printf("Semester %d SGPA = %.2f\n", semester, SGPA);

cumulative_TGC += semester_TGC;
cumulative_TC += semester_TC;
}

CGPA = (float)cumulative_TGC / cumulative_TC;

printf("\nGlobal ID = %d\n", global_id);
printf("Semesters = %d\n", m);

```

```

        printf("Cumulative
Credits = %d\n",
cumulative_TC);
        printf("CGPA = %.2f\n",
CGPA);

        return 0;
}

```

```

11
12 do {
13     printf("Enter Global ID: ");
14     scanf("%d", &global_id);
15     if (global_id <= 0)
16         printf("Invalid Global ID!\n");
17 } while (global_id <= 0);
18
19 do {
20     printf("Enter number of semesters: ");
21     scanf("%d", &m);
22     if (m <= 0)
23         printf("Invalid number of semesters!\n");
24 } while (m <= 0);
25
26 for (semester = 1; semester <= m; semester++)
27 {
28     semester_TC = 0;
29     semester_TGC = 0;
30
31     do {
32         printf("\nEnter number of courses for Semester %d: ", semester);
33         scanf("%d", &n);
34         if (n <= 0)
35             printf("Invalid number of courses!\n");
36     } while (n <= 0);
37
38     for (course = 1; course <= n; course++)
39     {
40         do {
41             printf("Enter course code: ");
42             scanf("%d", &code);
43             if (code < 0)

```

Invalid Global ID!
Enter Global ID: \$\$

Step - 5 Compilation and Execution

Case: 1

```
76         GP = 6;
77     else if (grade == 'E')
78         GP = 5;
79     else
80         GP = 4;
81
82     semester_TGC += GP * credit;
83     semester_TC += credit;
84 }
85
86 SGPA = (float)semester_TGC / semester_TC;
87 printf("Semester %d SGPA = %.2f\n", semester, SGPA);
88
89 cumulative_TGC += semester_TGC;
90 cumulative_TC += semester_TC;
91 }
92
93 CGPA = (float)cumulative_TGC / cumulative_TC;
94
95 printf("\nGlobal ID = %d\n", global_id);
96 printf("Semesters = %d\n", m);
```

Input

```
Enter Global ID: 101
Enter number of semesters: 1

Enter number of courses for Semester 1: 2
Enter course code: 102
Enter credit: 3,3
Enter grade (O/A/B/C/D/E/P): Invalid grade!
Enter grade (O/A/B/C/D/E/P): Invalid grade!
Enter grade (O/A/B/C/D/E/P): A,B
Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P): Invalid grade!
Enter grade (O/A/B/C/D/E/P): Semester 1 SGPA = 8.50

Global ID = 101
Semesters = 1
Cumulative Credits = 6
CGPA = 8.50
```

Case: 2

Case: 3

```
73         else if (grade == 'C')
74             GP = 7;
75         else if (grade == 'D')
76             GP = 6;
77         else if (grade == 'E')
78             GP = 5;
79         else
80             GP = 4;
81
82         semester_TGC += GP * credit;
83         semester_TC += credit;
84     }
85
86     SGPA = (float)semester_TGC / semester_TC;
87     printf("Semester %d SGPA = %.2f\n", semester, SGPA);
88
89     cumulative_TGC += semester_TGC;
90     cumulative_TC += semester_TC;
91 }
92
93 CGPA = (float)cumulative_TGC / cumulative_TC;
94
95 printf("\nGlobal ID = %d\n", global_id);
96 printf("Semesters = %d\n", m);
97 printf("Cumulative Credits = %d\n", cumulative_TC);
98 printf("CGPA = %.2f\n", CGPA);
99
100 return 0;
101 }
```

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input

Enter Global ID: 101

Enter number of semesters: 1

Enter number of courses for Semester 1: 0

Invalid number of courses!

Case: 4

```
74         GP = 7;
75         else if (grade == 'D')
76             GP = 6;
77         else if (grade == 'E')
78             GP = 5;
79         else
80             GP = 4;
81
82         semester_TGC += GP * credit;
83         semester_TC += credit;
84     }
85
86     SGPA = (float)semester_TGC / semester_TC;
87     printf("Semester %d SGPA = %.2f\n", semester, SGPA);
88
89     cumulative_TGC += semester_TGC;
90     cumulative_TC += semester_TC;
91 }
92
93 CGPA = (float)cumulative_TGC / cumulative_TC;
94
95 printf("\nGlobal ID = %d\n", global_id);
96 printf("Semesters = %d\n", m);
97 printf("Cumulative Credits = %d\n", cumulative_TC);
98 printf("CGPA = %.2f\n", CGPA);
99
100 return 0;
101 }
```

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input

Enter Global ID: 101

Enter number of semesters: 1

Enter number of courses for Semester 1: 2

Enter course code: 102

Enter credit: -2

Invalid credit!

Case: 5

```
76         GP = 6;
77     else if (grade == 'E')
78         GP = 5;
79     else
80         GP = 4;
81
82     semester_TGC += GP * credit;
83     semester_TC += credit;
84 }
85
86 SGPA = (float)semester_TGC / semester_TC;
87 printf("Semester %d SGPA = %.2f\n", semester, SGPA);
88
89 cumulative_TGC += semester_TGC;
90 cumulative_TC += semester_TC;
91 }
92
93 CGPA = (float)cumulative_TGC / cumulative_TC;
94
95 printf("\nGlobal ID = %d\n", global_id);
96 printf("Semesters = %d\n", m);
97 printf("Cumulative Credits = %d\n", cumulative_TC);
98 printf("CGPA = %.2f\n", CGPA);
99
100 return 0;
101 }
```

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Input

Enter Global ID: 101
Enter number of semesters: 1
Enter number of courses for Semester 1: 2
Enter course code: 102
Enter credit: 3,3
Enter grade (O/A/B/C/D/E/P): Invalid grade!
Enter grade (O/A/B/C/D/E/P): Invalid grade!
Enter grade (O/A/B/C/D/E/P): X
Invalid grade!

Case: 6

```
76         GP = 6;
77     else if (grade == 'E')
78         GP = 5;
79     else
80         GP = 4;
81
82     semester_TGC += GP * credit;
83     semester_TC += credit;
84 }
85
86 SGPA = (float)semester_TGC / semester_TC;
87 printf("Semester %d SGPA = %.2f\n", semester, SGPA);
88
89 cumulative_TGC += semester_TGC;
90 cumulative_TC += semester_TC;
91 }
92
93 CGPA = (float)cumulative_TGC / cumulative_TC;
94
95 printf("\nGlobal ID = %d\n", global_id);
96 printf("Semesters = %d\n", m);
97 printf("Cumulative Credits = %d\n", cumulative_TC);
98 printf("CGPA = %.2f\n", CGPA);
99
100 return 0;
101 }
```

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input

Semester 1 SGPA = 10.00

Global ID = 1
Semesters = 1
Cumulative Credits = 1
CGPA = 10.00

Step - 7 Conclusion

Thus, the C program for calculating SGPA and CGPA using nested loops and decision-making constructs was successfully implemented, compiled, executed, and tested using valid, invalid, and boundary test cases.